

Rito Chabalala

AI / Software Engineer | Full-Stack & Cloud Solutions | Data & AI Enablement

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Professional Summary

Junior AI / Software Engineer with 3+ years of hands-on experience delivering full-stack, cloud-based, and AI-driven solutions in enterprise environments. Proven ability to design, build, and deploy scalable applications using DevSecOps practices, modern cloud platforms, and AI technologies. Strong track record in developing production-ready systems, collaborating with cross-functional teams, and driving digitalisation and innovation initiatives.

Key Skills

Leadership & Strategy:

AI & Digital Enablement | Full-Stack & Cloud Architecture | Agile & Scrum Delivery | Stakeholder Engagement | Technical Governance & Best Practices | Data & AI Solutions | Problem Solving | Continuous Improvement

Technical Skills:

Languages: Python, R, Java, JavaScript, PHP, SQL, C++, HTML, CSS, LaTeX, Tailwind CSS

ML & AI: Machine Learning, Data Mining, Data Analysis, Data Engineering, Statistical Modelling, Predictive Analytics, AI, Generative AI (GenAI), LLMs, Retrieval-Augmented Generation (RAG), ML-based OCR

Libraries & Frameworks: Pandas, NumPy, Scikit-learn, TensorFlow, Matplotlib, LangChain, Django REST Framework, React (TypeScript),

Platforms & Tools: Microsoft Azure, Azure Kubernetes Service (AKS), Docker, Kubernetes, Azure DevOps, Git, GitHub, Power BI, Android Studio, Anaconda, Spyder

Databases: PostgreSQL, MySQL, MongoDB, Oracle

Productivity & Design: Microsoft Office (Advanced Excel, Word, PowerPoint), Visio, Adobe Photoshop, Illustrator, Lightroom

Other: CI/CD Pipelines, SDLC Best Practices, Synk, SonarQube, New Relic, Quality Assurance, Software Testing, Linux, Windows

PROFESSIONAL EXPERIENCE

Junior AI / Software Engineer | Sasol, Johannesburg

August 2024 – Present

- Lead end-to-end development of AI-driven and full-stack digital solutions, from architecture design to deployment and optimisation in production environments.
- Design and implement scalable AI and software systems aligned with enterprise architecture, cloud, and security best practices.
- Develop and maintain AI-enabled platforms, knowledge assistants, and data products to enhance decision-making and information accessibility.
- Build full-stack applications using React (TypeScript) and Django REST Framework, integrating AI/GenAI capabilities into enterprise solutions.
- Design and secure RESTful APIs and support cloud-native deployments on Azure Kubernetes Service (AKS).
- Apply DevSecOps, CI/CD, and cloud engineering practices using Azure DevOps, GitHub, Snyk, SonarQube, and New Relic.
- Contribute to the standardisation of AI architecture patterns and engineering practices across teams (South Africa and Germany).
- Collaborate with cross-functional stakeholders in agile environments to deliver data-driven solutions aligned with business objectives.
- Drive continuous improvement, innovation, and knowledge sharing supporting enterprise digitalisation and AI adoption initiatives.

Optimisation Developer / Data Scientist | Sasol, Johannesburg

April 2023 – July 2024

- Developed and deployed machine learning solutions to support operational optimisation and data-driven decision-making.
- Built predictive and statistical models using diverse datasets in collaboration with cross-functional stakeholders.
- Implemented ML-based OCR solutions to automate auditing processes, improving accuracy and compliance.
- Streamlined data pipelines and analytical workflows using Python-based tools.
- Designed and executed model validation and testing frameworks to ensure robustness and performance.
- Generated actionable insights from historical data, supporting business and operational improvements.

EDUCATION

2022 - 2023 **BSc Honours in Computer Science & Information Technology**
Sefako Makgatho Health Sciences University

Key Modules: Artificial Intelligence, Advanced Database Systems, Mobile Application Development, Health Informatics

Research Project: *Customers’ Perceptions Toward Mobile Banking Applications in Rural Areas of Limpopo*

2018 – 2021 **Bachelor of Science (BSc) – Mathematical Sciences**
Sefako Makgatho Health Sciences University

Majors: Computer Science, Statistics, Mathematics, Applied Mathematics

2017 - 2017 **Bachelor's Degree for NSC 2017**
Nkwangulatilo Education Centre

Subjects: Mathematics, Physical Sciences, Life Sciences, Agricultural Sciences, Xitsonga (Home Language), English (First Additional Language), Life Orientation

CERTIFICATIONS

Microsoft Certified: Security, Compliance, and Identity Fundamentals (SC-900) - 2025

National Certificate: Business Analysis (NQF Level 6, MICT SETA) - 2024

AWARDS & PROJECTS

Orbit - Innovation Management Platform (2026): Led the development of Innovation Orbit, an AI-enabled enterprise innovation management platform supporting idea generation and collaboration.

AI Knowledge Assistant (RAG) (2025): Developed a retrieval-augmented generation system integrating vector search and LLMs, including document ingestion, embeddings, and grounding validation.

Full-Stack Blog Application (2024): Built a Django REST and React-based platform with authentication and full CRUD functionality.

Huawei Seeds for the Future (2023): Selected participant; part of a team awarded Top Project in Africa for innovation and technical excellence.

Disease Prediction System (2023): Developed a machine-learning model to predict diseases from user-reported symptoms using data preprocessing and predictive modelling techniques.

University Online Admission System (2022): Designed and implemented an automated admissions system to improve efficiency and reduce manual processing.

E-Commerce Platform (2021): Developed a full-stack web application supporting multi-brand product browsing and online transactions.

REFERENCES

References are available on request.